

Sipan Bareyan

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EDUCATION

Paris-Saclay University, Faculty of Sciences of Orsay

Orsay, France

Double Bachelor's Degree in Computer Science and Mathematics

Sept. 2023 – June 2026

- Relevant coursework: Probability, Integration, Differential Calculus and Optimization, Linear Algebra (I-II), Advanced Algorithms, Data Structures and Algorithms, Statistical Learning, Numerical Analysis, Modeling and Simulation

TUMO Center for Creative Technologies

Yerevan, Armenia

Technology and Design Program

2015 – 2019

- Courses taken: Programming, Game Development, 3D Modeling, Web Development, Robotics

Anania Shirakatsy Lyceum

Yerevan, Armenia

Secondary Education

Sept. 2011 – May 2023

Overall GPA: 3.945/4

EXPERIENCE

Research Intern – Numerical Methods

May 2026 – Jun. 2026

Université Paris-Saclay

Orsay, France

- Investigated automatic differentiation techniques for nonlinear differential equation solvers.
- Compared methods for gradient computation in PDE-constrained optimization problems.
- Implemented spectral PDE solvers, adjoint methods, and custom automatic differentiation rules in Julia.
- github.com/bareyan/PDE_adjoint_solver

Student Assistant

September 2025 –

Lumen Library – Paris-Saclay University

Gif-sur-Yvette, France

- Part-time work within the university library

Intern – LAHDAK Team

June 2025 – July 2025

LISN (Interdisciplinary Laboratory of Digital Sciences)

Gif-sur-Yvette, France

- Worked on heterogeneous knowledge graphs
- Prepared data for training machine learning models
- Developed a knowledge graph exploration tool with automatic data cleaning based on statistical methods
- <https://github.com/bareyan/rust-kg-explorer>

Front-end Web Development Intern

April 2023 – May 2023

Volo

Yerevan, Armenia

- Developed dynamic and responsive web applications with React

Freelance Developer (Web & Mobile)

August 2020 – August 2023

Remote

- Developed websites and web applications for startups and companies.
- Developed mobile applications with Dart/Flutter for international clients.
- Kwork.ru profile: https://kwork.ru/user/sb_dev

PROJECTS

EduViz | (*in development*)

Devpost

- Prototype that transforms documents/notes into visual explanations (videos) with an educational structure.
- Examples: youtube.com/playlist?list=PLu00ypjWn09trvhdCKQJ0c-5pb90uda-9

Rust KG Explorer

github.com/bareyan/rust-kg-explorer

- Knowledge graph explorer: entity navigation and SPARQL queries through a local interface.
- Import of datasets (Web Data Commons) or RDF files; tool used during my internship at LISN.
- Automatic cleaning and processing of heterogeneous graphs.

Tetris AI (university project)

Report

- Implementation of Tetris and study of polyominoes (transformations, classes, generation) documented in a report.
- Heuristic AI with weights learned through a genetic algorithm (3,000–10,000 reported lines).

LaNoteX | (*in development*)

LinkedIn Post

- Student application for organizing courses and generating study materials from imported content.
- Example repository: github.com/bareyan/lecture_notes.

PARTICIPATION IN COMPETITIONS AND EVENTS

JFP 15 (15th Paris Region Programming Day) – 2nd place, Paris, France

Yeghiazaryan Cup 2021 & 2022 – Programming competition, Yerevan, Armenia

Independence Cup 2021 – Programming competition, Yerevan, Armenia (special prize)

Robotex 2019 – International robotics competition, Yerevan, Armenia (smart city robot)

Digitech Expo 2019 – Yerevan, Armenia (presentation of a mobile drone project)

VOLUNTEERING

Volunteer at Anania Shirakatsy Lyceum, Yerevan, Armenia — Jan. 2019 – May 2023

Volunteer at CEPFA, Yerevan, Armenia

TECHNICAL SKILLS

Programming Languages : Python, C, C++, Java, Rust, JavaScript, Dart, OCaml, Processing

Query Languages : SQL, SPARQL

Frameworks : React, Flutter

Tools : Linux, Git, Blender

Markup Languages and Data Formats : LaTeX, HTML & CSS, RDF, RDFa

Libraries : NumPy, Matplotlib, PyTorch, SciPy, Scikit-learn

LANGUAGES

Armenian (native), English (advanced – C1), French (professional proficiency), Russian (advanced)

STANDARDIZED TESTS

TOEIC (March 2026): 990 — SAT (March 2022) Mathematics: 780 — DELF B2 obtained (November 2022)